PRATIKUM MODUL 2

PENGEMBANGAN UI LANJUT

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Struktur Aplikasi dan Navigasi Dalam Vue.Js

Week 2

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PROGRAM STUDI SISTEM INFORMASI

FAKULTAS REKAYASA INDUSTRI

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1. TUJUAN

   Memahami serta menerapkan keterkaitan Site Map, Views, dan Router, membangun hierarki halaman Parent Child merancang transisi navigasi interaktif Navbar dengan routerlink serta membuat penunjuk posisi (Breadcrumb) pada aplikasi web.

2. ALAT DAN BAHAN
   a. Komputer atau Laptop
   b. Visual Studio Code
   c. Vue.js
   d. Node.js

3. KODE DAN PENJELASAN

A. Inisialisasi dan Kerangka Dasar

a. src/main.js

Merupakan entry point utama aplikasi Vue. File ini bertugas membuat instance aplikasi, mendaftarkan modul state management dan sistem navigasi.

belajar-vue > src > JS main.js > ...

```
1   import './assets/main.css'
2
3   import { createApp } from 'vue'
4   import { createPinia } from 'pinia'
5
6   import App from './App.vue'
7   import router from './router'
8
9   const app = createApp(App)
10
11  app.use(createPinia())
12  app.use(router)
13
14  app.mount('#app')
```

b. src/App.vue

Berfungsi sebagai root component dasar. Berisi elemen yang bertindak sebagai placeholder untuk menampilkan komponen halaman sesuai dengan rute URL yang sedang aktif.

belajar-vue > src > V App.vue > ...

```
1   <script setup>
2   import { RouterView } from 'vue-router'
3   </script>
4
5   <template>
6     <RouterView />
7   </template>
```

c. src/layouts/App.vue

Sebagai layout global aplikasi. Menggabungkan komponen navigasi atas (Navbar), penunjuk posisi (Breadcrumb), dan di dalam kontainer utama berformat responsif agar struktur navigasi tetap konsisten di setiap halaman.



B. Mengatur Rute

a. src/router/index.js

Menginisialisasi Vue Router menggunakan createWebHistory dan mendefinisikan pemetaan URL ke komponen tertentu (Home, About, Contact, Browse).



C. Transisi dan Orientasi (Navbar & Breadcrumb)

a. src/components/app/Navbar.vue

Mengelola navigasi utama sebagai instrumen transisi antarhalaman.



```vue
<script setup>
import { ref, onMounted, onUnmounted } from 'vue'
import { useRoute } from 'vue-router'

const isScrolled = ref(false)
const route = useRoute()

const menus = [
  { name: 'Home', path: '/' },
  { name: 'About', path: '/about' },
  {
    name: 'Browse',
    path: '/browse',
    children: [
      {
        name: 'Event List',
        path: '/browse/events',
        children: [
          {
            name: 'Event Detail (Sample)',
            path: '/browse/events/1'
          }
        ]
      },
      {
        name: 'Category',
        path: '/browse/category'
      }
    ]
  },
  { name: 'Contact', path: '/contact' }
]

const handleScroll = () => {
  isScrolled.value = window.scrollY > 10
}

onMounted(() => {
  window.addEventListener('scroll', handleScroll)
})

onUnmounted(() => {
  window.removeEventListener('scroll', handleScroll)
})
</script>

<template>
  <nav class="navbar" :class="{ scrolled: isScrolled }">
    <div class="navbar-container">

      <!-- Logo -->
      <router-link to="/" class="logo">
        <svg
          width="28"
          height="32"
          viewBox="0 0 24 28"
          fill="none"
          xmlns="http://www.w3.org/2000/svg"
          class="logo-icon"
        >
```

```
        <path
          d="M12 0L22.3923 6V18L12 24L1.6077 18V6L12 0Z"
          fill="white"
        />

        <path
          d="M15 9.5 C15 9.5 14 8 12 8 C9.5 8 8 10 8 12.5 C8 15 9.5 17 12 17 C14 17 15 16 15.5 14.5 V 12.5 H 12.5"
          stroke="#1A1643"
          stroke-width="2.5"
          stroke-linecap="round"
          stroke-linejoin="round"
        />
      </svg>

      <span class="logo-text">Gatherly</span>
    </router-link>

    <!-- Navigation Menu -->
    <ul class="nav-menu">
      <li
        v-for="menu in menus"
        :key="menu.name"
        class="nav-item"
      >
        <router-link
          :to="menu.path"
          class="nav-link"
          :class="{
            active:
              route.path === menu.path ||
              (menu.path !== '/' && route.path.startsWith(menu.path))
          }"
        >
          {{ menu.name }}

          <svg
            v-if="menu.children"
            class="dropdown-indicator"
            width="12"
            height="12"
            viewBox="0 0 24 24"
            fill="none"
            stroke="currentColor"
            stroke-width="2"
            stroke-linecap="round"
            stroke-linejoin="round"
          >
            <polyline points="6 9 12 15 18 9" />
          </svg>
        </router-link>
```

```
        </router-link>

        <!-- First Level Dropdown -->
        <ul v-if="menu.children" class="dropdown-menu">
          <li
            v-for="child in menu.children"
            :key="child.name"
            class="dropdown-item"
          >
            <router-link
              :to="child.path"
              class="dropdown-link"
            >
              {{ child.name }}

              <svg
                v-if="child.children"
                class="submenu-indicator"
                width="12"
                height="12"
                viewBox="0 0 24 24"
                fill="none"
                stroke="currentColor"
                stroke-width="2"
                stroke-linecap="round"
                stroke-linejoin="round"
              >
                <polyline points="9 18 15 12 9 6" />
              </svg>
            </router-link>

            <!-- Second Level Dropdown -->
            <ul v-if="child.children" class="submenu">
              <li
                v-for="subchild in child.children"
                :key="subchild.name"
                class="submenu-item"
              >
                <router-link
                  :to="subchild.path"
                  class="dropdown-link"
                >
                  {{ subchild.name }}
                </router-link>
              </li>
            </ul>
          </li>
        </ul>
      </li>
    </ul>
```

```
    <!-- Right Navigation -->
    <div class="nav-right">

      <!-- Language Selector -->
      <div class="lang-selector">
        <svg
          width="20"
          height="20"
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
          stroke-linecap="round"
          stroke-linejoin="round"
        >
          <circle
            cx="12"
            cy="12"
            r="10"
          />

          <line
            x1="2"
            y1="12"
            x2="22"
            y2="12"
          />

          <path
            d="M12 2a15.3 15.3 0 0 1 4 10 15.3 15.3 0 0 1-4 10 15.3 15.3 0 0 1-4-10 15.3 15.3 0 0 1 4-10z"
          />
        </svg>

        <span class="lang-text">EN</span>

        <svg
          width="18"
          height="18"
          viewBox="0 0 24 24"
          fill="none"
          stroke="currentColor"
          stroke-width="2"
          stroke-linecap="round"
          stroke-linejoin="round"
          class="chevron"
        >
          <polyline points="6 9 12 15 18 9" />
        </svg>
      </div>

```

```
        <!-- Hamburger -->
        <button class="hamburger-btn">
          <svg
            width="24"
            height="24"
            viewBox="0 0 24 24"
            fill="none"
            stroke="currentColor"
            stroke-width="1.5"
            stroke-linecap="round"
            stroke-linejoin="round"
          >
            <line x1="4" y1="12" x2="20" y2="12"></line>
            <line x1="4" y1="6" x2="20"y2="6"></line>
            <line x1="4" y1="18"x2="20" y2="18"></line>
          </svg>
        </button>

      </div>
    </div>
  </nav>
</template>
```

b. src/components/app/Breadcrumb.vue

Berfungsi sebagai instrumen orientasi pengguna (location indicator). Mengubah parameter dinamis pada rute detail menjadi URL penuh serta memastikan rute dasar (Home) selalu berada di posisi terdepan. Menampilkan tautan aktif dan pemisah (/) secara visual di antarmuka.



```
belajar-vue > src > components > app > Breadcrumb.vue > ...
<script setup>
import { computed } from 'vue';
import { useRoute } from 'vue-router';

const route = useRoute();

const breadcrumbs = computed(() => {
  const matched = route.matched;
  let crumbs = matched.map((m) => {
    let path = m.path;
    if (path.includes(':')) {
      path = route.path;
    }
    return {
      name: m.name,
      path: path,
      meta: m.meta
    };
  }).filter(m => m.meta && m.meta.breadcrumb);

  if (route.name === 'event-detail') {
    crumbs.splice(crumbs.length - 1, 0, {
      path: '/browse/events',
      meta: { breadcrumb: 'Event List' }
    });
  }

  // Prepend Beranda if it's not already the first item
  if (crumbs.length === 0 || crumbs[0].meta.breadcrumb !== 'Home') {
    crumbs.unshift({
      path: '/',
      meta: { breadcrumb: 'Home' }
    });
  }

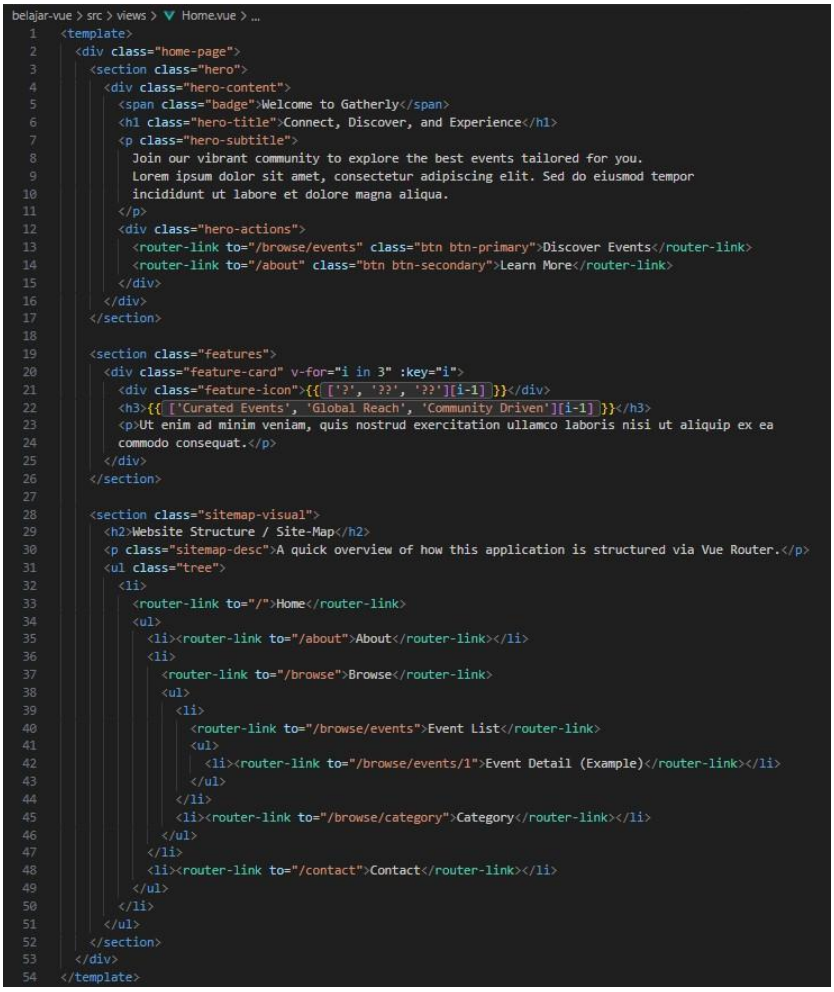
  return crumbs;
});
</script>

<template>
  <nav class="breadcrumb" v-if="breadcrumbs.length > 0">
    <ul>
      <li v-for="(crumb, index) in breadcrumbs" :key="index">
        <span v-if="index > 0" class="separator">/</span>
        <router-link v-if="index < breadcrumbs.length - 1" :to="crumb.path">
          {{ crumb.meta.breadcrumb }}
        </router-link>
        <span v-else class="active-crumb">{{ crumb.meta.breadcrumb }}</span>
      </li>
    </ul>
  </nav>
</template>
```

D. Destinasi Halaman Pendukung

a. src/views/Home.vue

Merupakan halaman awal yang memuat hero section, ringkasan fitur utama, serta visualisasi struktur situs untuk menggambarkan arsitektur rute Vue Router.



```
belajar-vue > src > views > Home.vue > ...
<template>
  <div class="home-page">
    <section class="hero">
      <div class="hero-content">
        <span class="badge">Welcome to Gatherly</span>
        <h1 class="hero-title">Connect, Discover, and Experience</h1>
        <p class="hero-subtitle">
          Join our vibrant community to explore the best events tailored for you.
          Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor
          incididunt ut labore et dolore magna aliqua.
        </p>
        <div class="hero-actions">
          <router-link to="/browse/events" class="btn btn-primary">Discover Events</router-link>
          <router-link to="/about" class="btn btn-secondary">Learn More</router-link>
        </div>
      </div>
    </section>

    <section class="features">
      <div class="feature-card" v-for="i in 3" :key="i">
        <div class="feature-icon">{{ ['?', '??', '??'][i-1] }}</div>
        <h3>{{ ['Curated Events', 'Global Reach', 'Community Driven'][i-1] }}</h3>
        <p>Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea
        commodo consequat.</p>
      </div>
    </section>

    <section class="sitemap-visual">
      <h2>Website Structure / Site-Map</h2>
      <p class="sitemap-desc">A quick overview of how this application is structured via Vue Router.</p>
      <ul class="tree">
        <li>
          <router-link to="/">Home</router-link>
          <ul>
            <li><router-link to="/about">About</router-link></li>
            <li>
              <router-link to="/browse">Browse</router-link>
              <ul>
                <li>
                  <router-link to="/browse/events">Event List</router-link>
                  <ul>
                    <li><router-link to="/browse/events/1">Event Detail (Example)</router-link></li>
                  </ul>
                </li>
                <li><router-link to="/browse/category">Category</router-link></li>
              </ul>
            </li>
            <li><router-link to="/contact">Contact</router-link></li>
          </ul>
        </li>
      </ul>
    </section>
  </div>
</template>
```

b. src/views/About.vue

Merepresentasikan perpindahan antarmenu utama pada tingkat hierarki yang sama. About.vue berisi profil, visi, dan misi platform.

belajar-vue > src > views > About.vue > ...

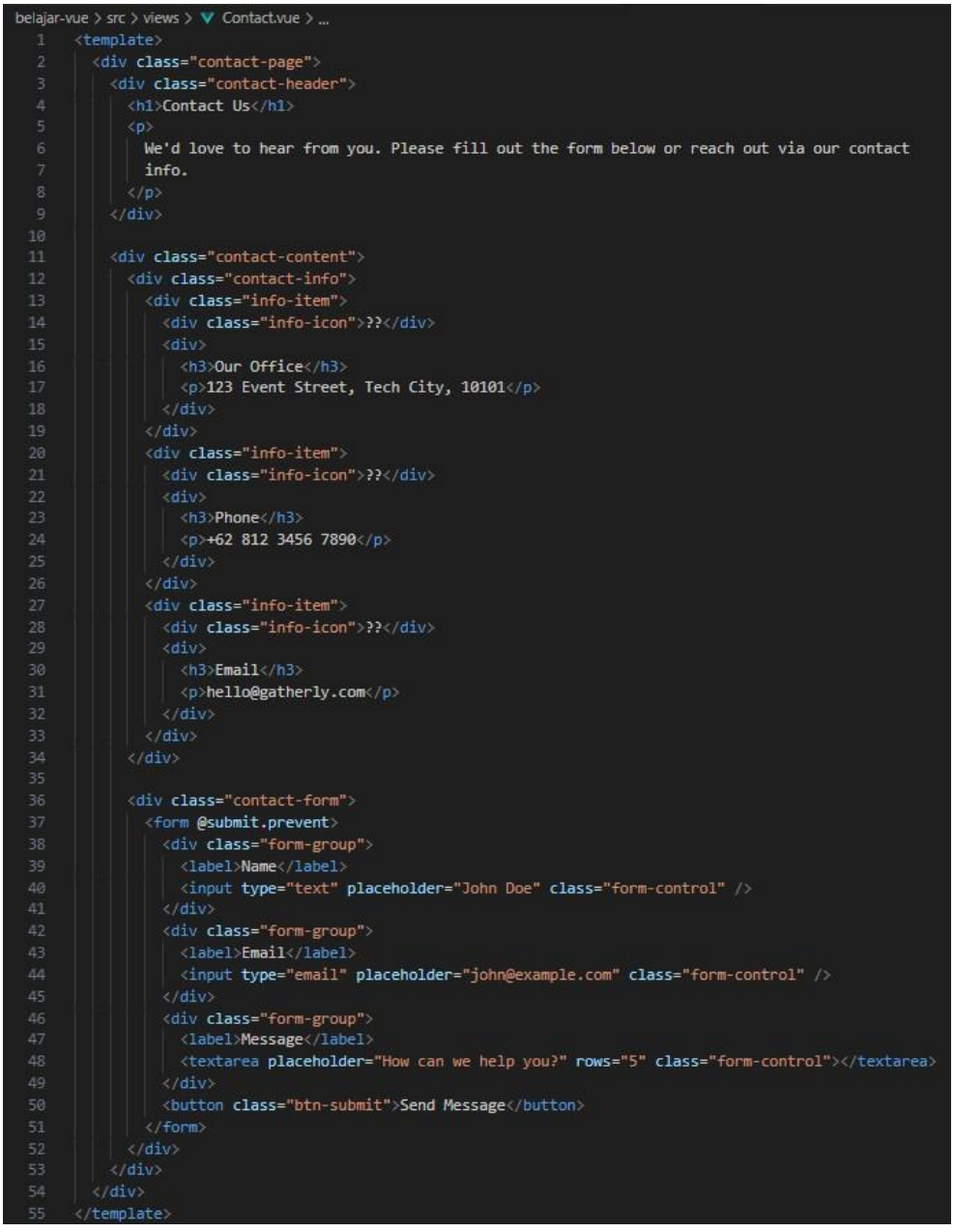
```vue
<template>
  <div class="about-page">
    <div class="about-header">
      <h1>About Gatherly</h1>
      <p>Building the future of event experiences, one community at a time.</p>
    </div>

    <div class="about-content">
      <div class="text-block">
        <h2>Our Mission</h2>
        <p>
          Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat
          nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia
          deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis iste natus error sit
          voluptatem accusantium doloremque laudantium.
        </p>
      </div>

      <div class="text-block">
        <h2>Our Vision</h2>
        <p>
          Lorem ipsum dolor sit amet, consectetur adipiscing elit. Integer nec odio. Praesent
          libero. Sed cursus ante dapibus diam. Sed nisi. Nulla quis sem at nibh elementum
          imperdiet. Duis sagittis ipsum. Praesent mauris. Fusce nec tellus sed augue semper porta.
        </p>
      </div>
    </div>
  </div>
</template>
```

c. src/views/Contact.vue

Merepresentasikan perpindahan antarmenu utama pada tingkat hierarki yang sama. sedangkan Contact.vue menyediakan informasi kontak dan formulir pengiriman pesan.



belajar-vue > src > views > Contact.vue > ...

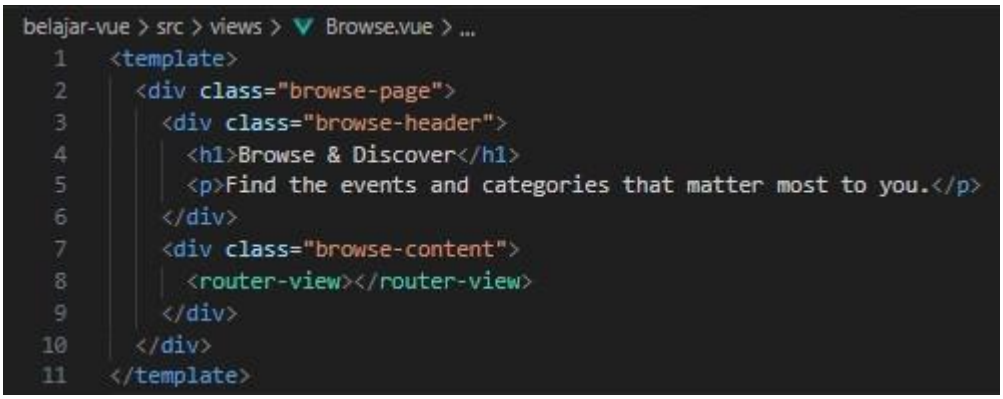
```
<template>
  <div class="contact-page">
    <div class="contact-header">
      <h1>Contact Us</h1>
      <p>
        We'd love to hear from you. Please fill out the form below or reach out via our contact
        info.
      </p>
    </div>

    <div class="contact-content">
      <div class="contact-info">
        <div class="info-item">
          <div class="info-icon">??</div>
          <div>
            <h3>Our Office</h3>
            <p>123 Event Street, Tech City, 10101</p>
          </div>
        </div>
        <div class="info-item">
          <div class="info-icon">??</div>
          <div>
            <h3>Phone</h3>
            <p>+62 812 3456 7890</p>
          </div>
        </div>
        <div class="info-item">
          <div class="info-icon">??</div>
          <div>
            <h3>Email</h3>
            <p>hello@gatherly.com</p>
          </div>
        </div>
      </div>

      <div class="contact-form">
        <form @submit.prevent>
          <div class="form-group">
            <label>Name</label>
            <input type="text" placeholder="John Doe" class="form-control" />
          </div>
          <div class="form-group">
            <label>Email</label>
            <input type="email" placeholder="john@example.com" class="form-control" />
          </div>
          <div class="form-group">
            <label>Message</label>
            <textarea placeholder="How can we help you?" rows="5" class="form-control"></textarea>
          </div>
          <button class="btn-submit">Send Message</button>
        </form>
      </div>
    </div>
  </div>
</template>
```

d. src/views/Browse.vue

Merupakan komponen wrapper untuk route untuk menampilkan halaman turunan child routes tanpa mengubah kerangka utama.



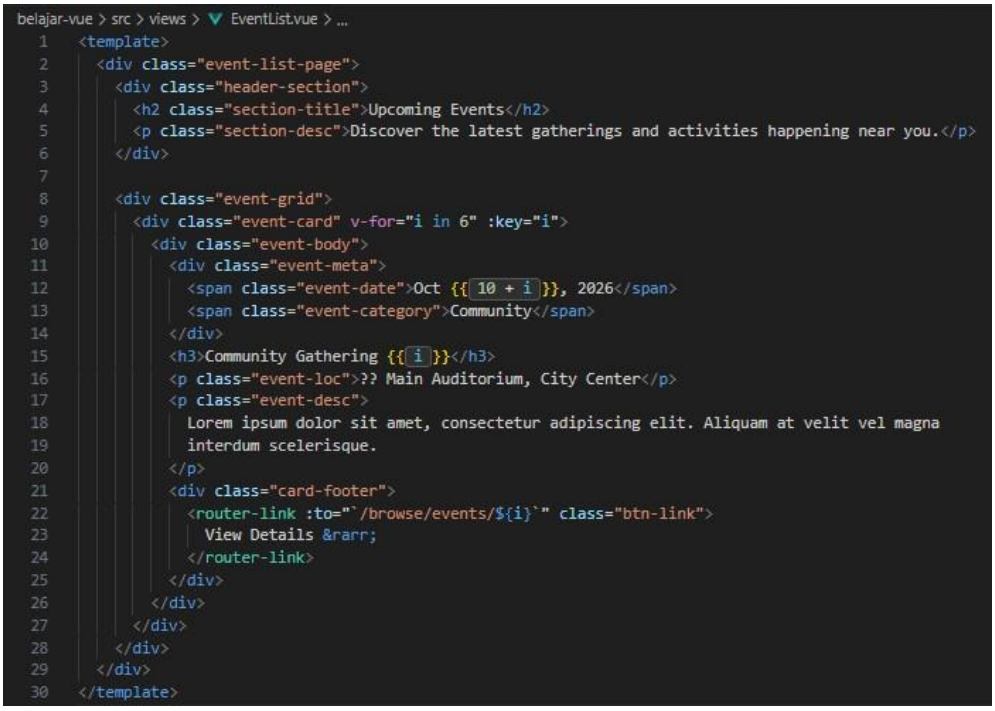
e. src/views/Category.vue

Merupakan halaman turunan dari Browse yang menampilkan klasifikasi acara berdasarkan kategori menggunakan tata letak berbasis grid.



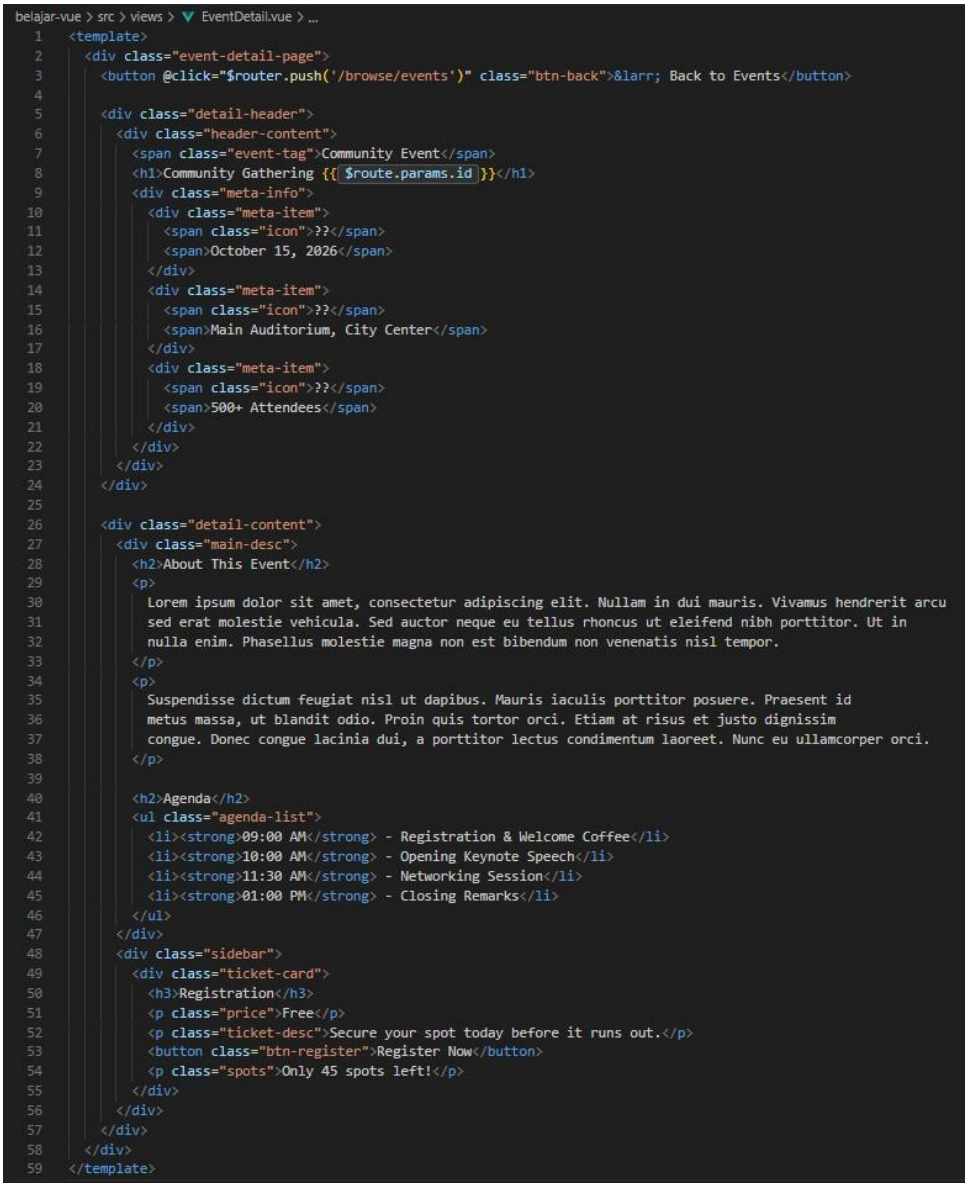
f. src/views/EventList.vue

Menampilkan daftar katalog acara dan menyediakan tautan rute dinamis menuju halaman rincian masing-masing acara.



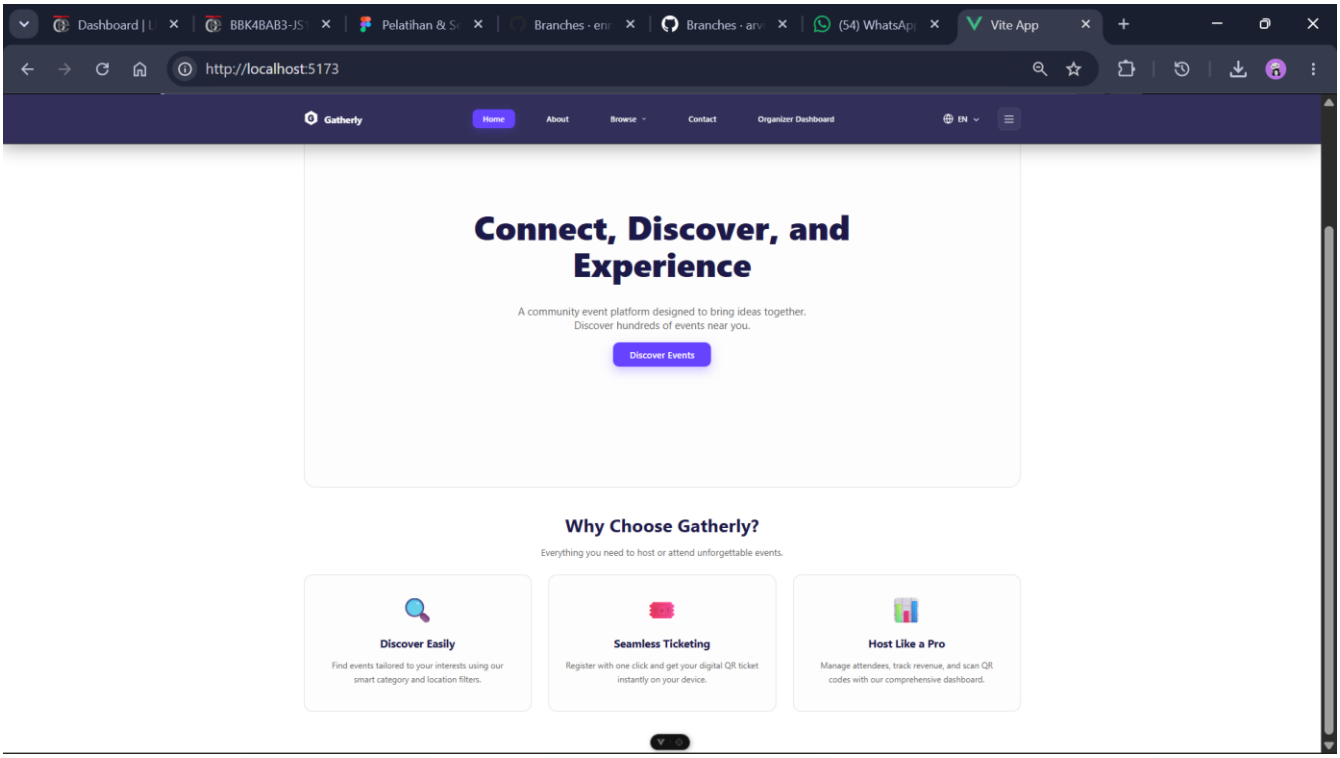
4. src/views/EventDetail.vue

Merupakan tingkat terdalam navigasi yang menampilkan data berdasarkan parameter URL dinamis. Halaman ini juga menyediakan navigasi kembali menggunakan serta menampilkan rincian agenda dan registrasi tiket.

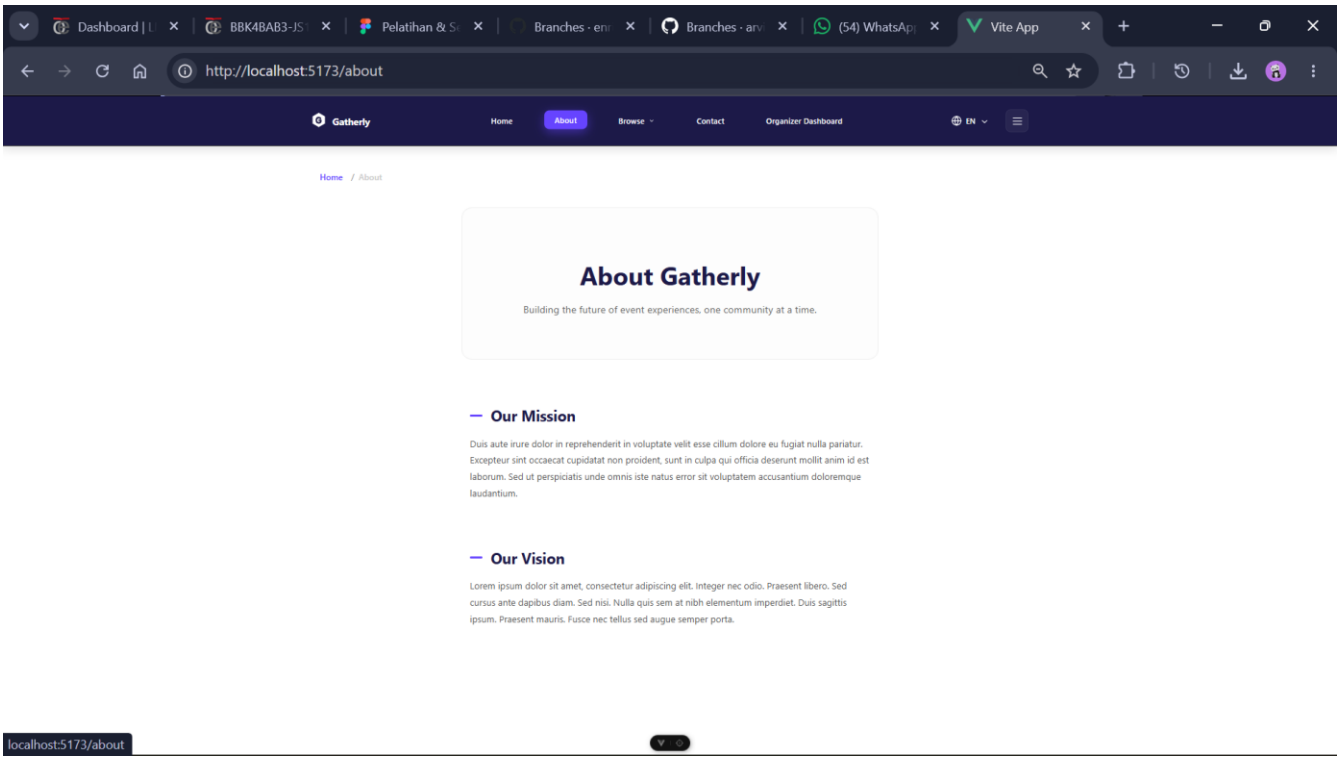


4. SCREENSHOT HASIL PRAKTIK

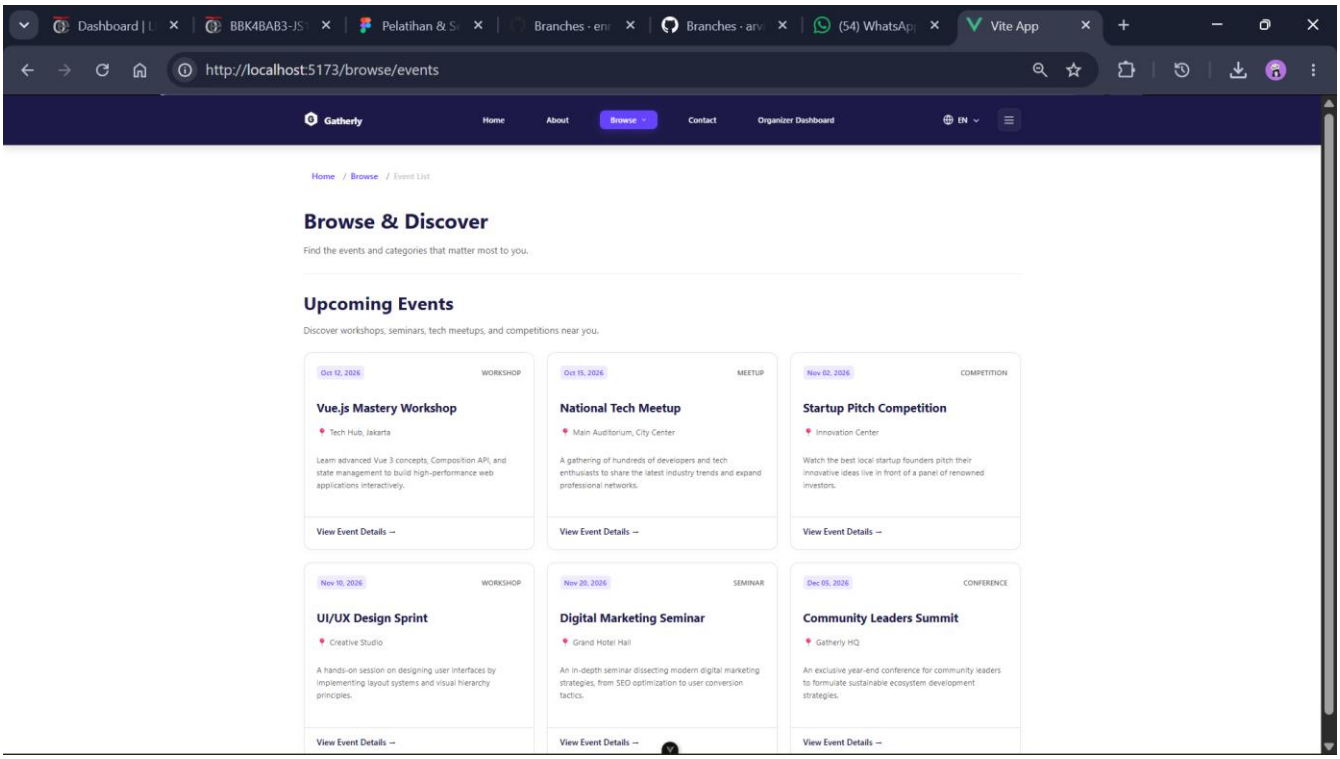
a. Tampilan home



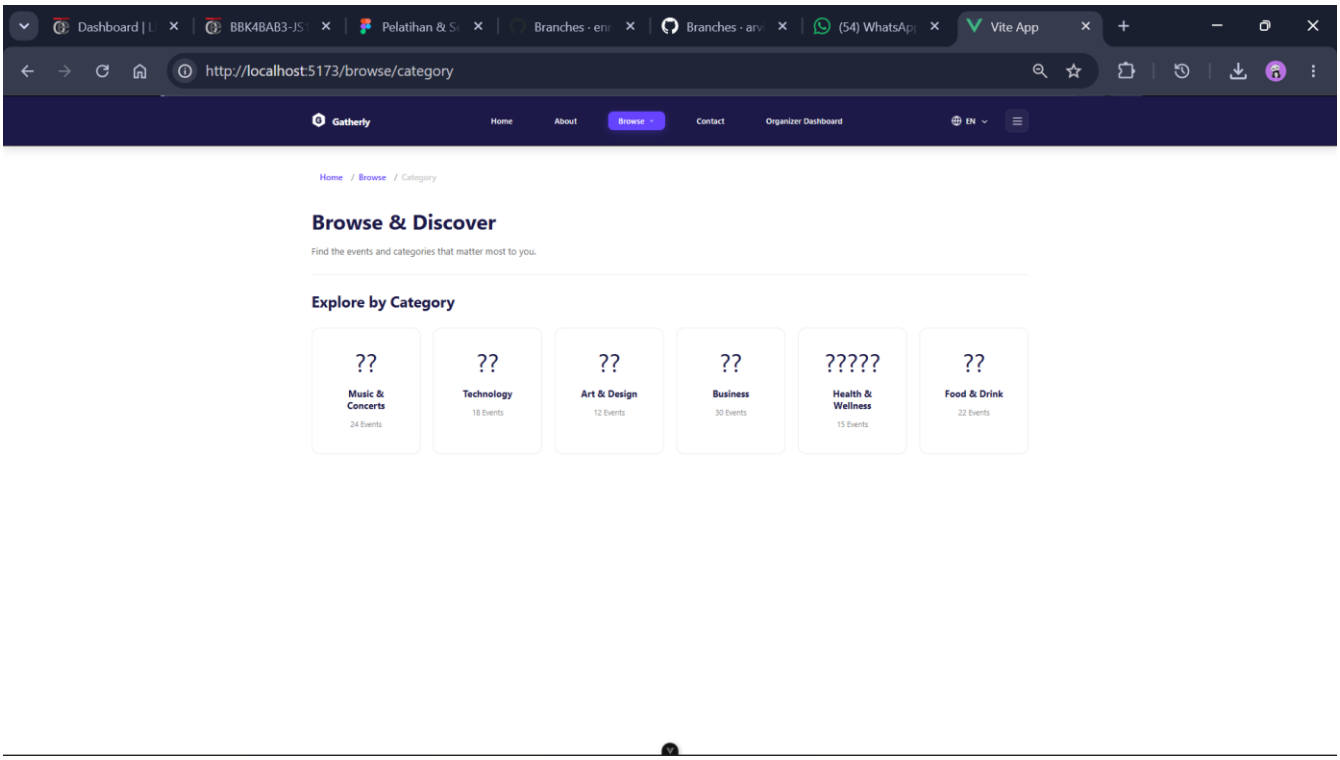
b. /about



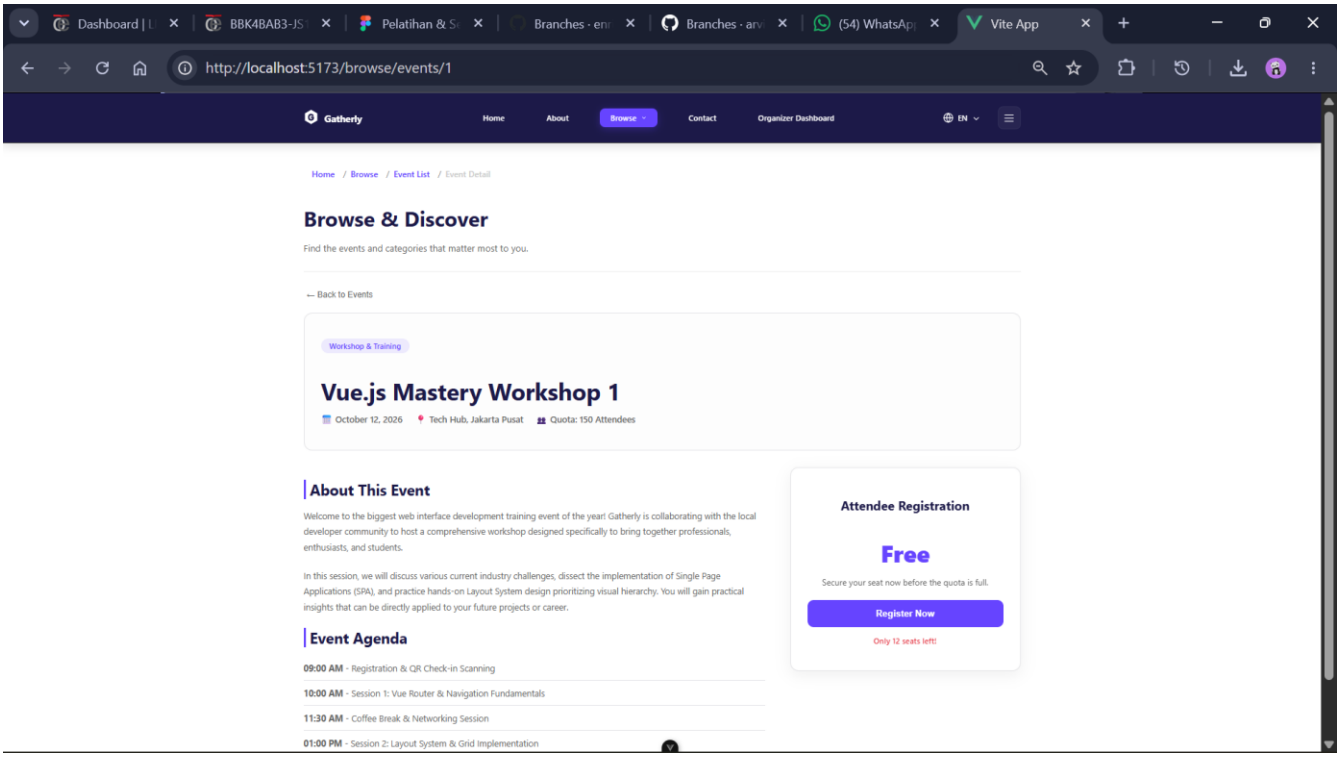
c. /browse/events



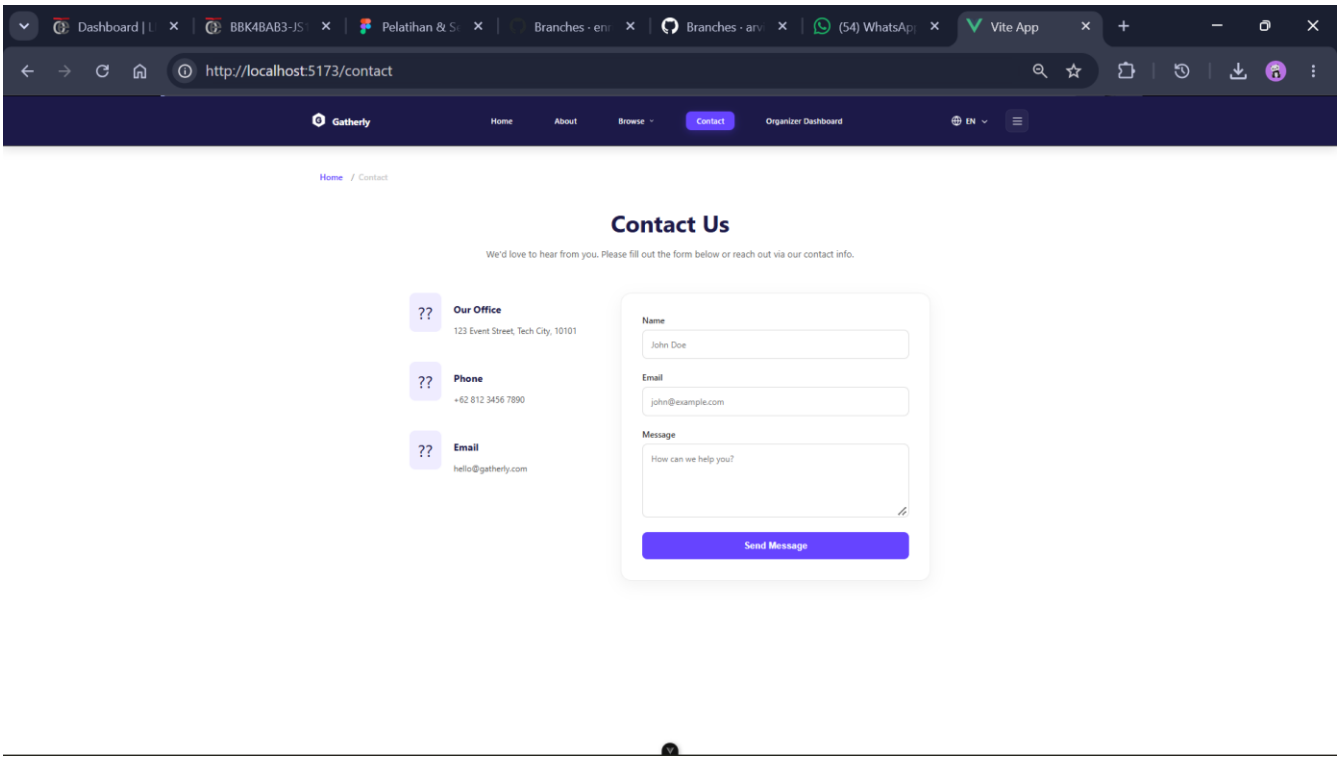
d. /browse/category



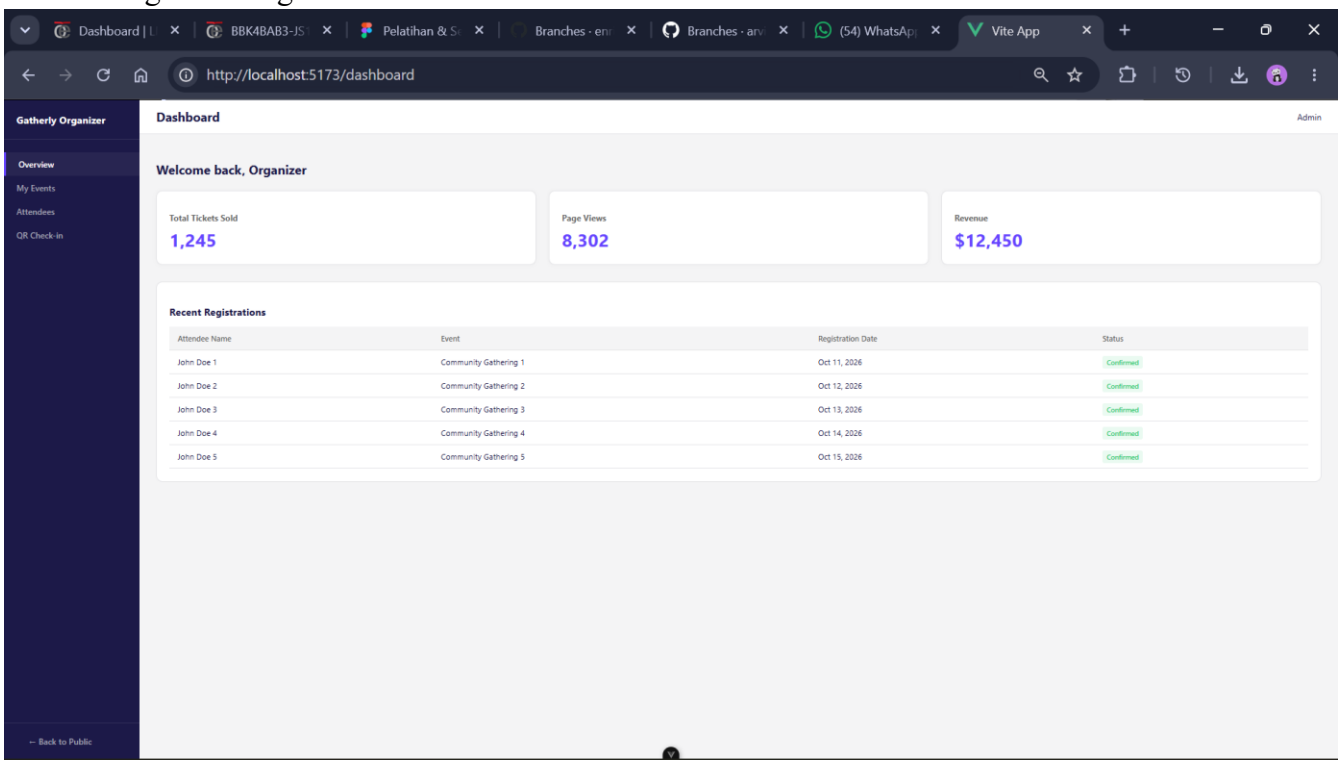
e. /browse/events/1



f. /contact



g. Organizer Dashboard



5. KESIMPULAN

Berdasarkan praktikum yang telah dilaksanakan, dapat disimpulkan bahwa penerapan arsitektur aplikasi menggunakan Vue.js berhasil mengoptimalkan pengelolaan struktur website serta alur navigasi antarmuka tanpa memerlukan pemuatan ulang halaman secara penuh. Integrasi hierarki rute parent child dan dynamic route yang dipadukan dengan komponen Navbar sebagai instrumen transisi serta Breadcrumb sebagai instrumen orientasi terbukti mampu menciptakan alur pergerakan pengguna secara terstruktur, efisien, dan responsif sesuai standar perancangan antarmuka modern.

6. LINK GITHUB

https://github.com/arviamartha/Belajar_Vue.js/tree/modul-2